

Gabriel Martinez

WES 237B

Assignment 3

Assignment 3 Report

Lab

The very first thing I did was get more information about SIMD [1]. Within the research phase I found a very good video that explained neon intrinsics [2] specifically helped me with understanding the naming convention of, e.g., `<type><size>x<#lanes>_t`. So, armed with this knowledge I was very quickly able to lookup the appropriate neon intrinsics functions and vector data type for the lab implementation [3].

1. How do we handle when there is not at least four elements remaining?
 - a. Answer: Keep track of the index and have a separate for loop to iterate through the remaining elements.

```
44     int elementCount = rows * cols;
45     //int vectorEnd = elementCount - (elementCount % 4);
46     int sum = 0;
47     int i = 0;
48
49     // Process four integers per iteration
50     for (; i + 4 < elementCount; i += 4) {
51         // Load 4 32 bit integers into a 128bit NEON vector
52         int32x4_t values = vld1q_s32(&host_a.data[i]);
53         sum += vaddvq_s32(values);
54     }
55
56     // Process the remaining integers
57     for (; i < elementCount; ++i) {
58         sum += host_a.data[i];
59     }
60 }
```

2. What is happening at the hardware level that may make this more efficient?
 - a. Answer: The NEON vector instruction set extensions for ARM provide Single Instruction Multiple Data (SIMD) capabilities [4]. At the hardware level, NEON allows the CPU to process four integers at once instead of one at a time. This reduces the number of instructions and may make the program faster by enabling data level parallelism.

Lab Output

```
ubuntu@c28a4dfb50ef:/workspaces/wes237b/assignment-3/lab/neon$ make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -I/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -I/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -I/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -I/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -l/usr/local/cuda/include -l../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/output.raw output.raw
sum: 129 == 129
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/output.raw output.raw
sum: -11 == -11
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/output.raw output.raw
sum: 143 == 143
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/output.raw output.raw
sum: 305 == 305
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/output.raw output.raw
sum: -270 == -270
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/output.raw output.raw
sum: -46 == -46
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/output.raw output.raw
sum: 986 == 986
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/output.raw output.raw
sum: 760 == 760
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/output.raw output.raw
sum: 76 == 76
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/output.raw output.raw
sum: 778 == 778
!!SOLUTION IS CORRECT!!
ubuntu@c28a4dfb50ef:/workspaces/wes237b/assignment-3/lab/neon$
```

Assignment 3 Report

3.1 OpenCL Naïve Matrix Multiply

Running `time make run` produced the following results:

Measurement	Assignment #2	Assignment #3	Speedup (x)
Real time	2m 55.721s	0m 50.718s	3.46x
User time	2m 54.587s	0m 2.154s	
System time	0m 0.254s	0m 0.755s	

Assignment 3 was approximately 3.46x faster overall based on the real execution time. However, timing each dataset separately revealed that Assignment 2 was faster for smaller matrices, such as Dataset/0, while Assignment 3 performed better as the matrix sizes increased.

3.1.2 Naïve Matrix Multiply Output (`time make run`)

```
ubuntu@ubuntu2044dfb58ef:/workspaces/wes237b/assignment-3/homework/opencv_naive_mm$ time make run
cd .././helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar res helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution .././helper_lib/helper_lib.a main.c -I.././helper_lib -l/usr/local/cuda/include -L.././helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m50.718s
user    0m2.154s
sys     0m0.755s
ubuntu@ubuntu2044dfb58ef:/workspaces/wes237b/assignment-3/homework/opencv_naive_mm$
```

3.1.3 Naïve Matrix Multiply Output (`make benchmark`)

I modified the Makefile to run `time` for each dataset folder.

```

• ubuntu@c2ba4dfb59ef: /workspaces/wss237b/assignment-3/homework/opencv_naive_m$ make benchmark
time ./solution Dataset/8/input8.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.182s
user    0m0.044s
sys     0m0.028s
time ./solution Dataset/1/input8.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.183s
user    0m0.043s
sys     0m0.053s
time ./solution Dataset/2/input8.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.113s
user    0m0.046s
sys     0m0.028s
time ./solution Dataset/3/input8.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.185s
user    0m0.042s
sys     0m0.044s
time ./solution Dataset/4/input8.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.888s
user    0m0.039s
sys     0m0.043s
time ./solution Dataset/5/input8.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.184s
user    0m0.042s
sys     0m0.047s
time ./solution Dataset/6/input8.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.886s
user    0m0.048s
sys     0m0.028s
time ./solution Dataset/7/input8.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.877s
user    0m0.036s
sys     0m0.035s
time ./solution Dataset/8/input8.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.882s
user    0m0.037s
sys     0m0.038s
time ./solution Dataset/9/input8.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m0.895s
user    0m0.054s
sys     0m0.036s
time ./solution Dataset/10/input8.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m09.165s
user    0m1.452s
sys     0m0.159s
• ubuntu@c2ba4dfb59ef: /workspaces/wss237b/assignment-3/homework/opencv_naive_m$

```

3.2 OpenCL Tiled Matrix Multiplication

Measurement	3.1 OpenCL Naïve MM	3.2 OpenCL Tiled MM	Speedup (x)
Real time	0m 50.718s	0m4.310s	11.77x
User time	0m 2.154s	0m2.258s	
System time	0m 0.755s	0m0.785s	

The tiled OpenCL matrix multiplication achieved an 11.77x speedup over the naive OpenCL implementation.

3.2.1 OpenCL Tiled Matrix Multiplication Output (`time make run`)

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/opencl_tiled_m$ time make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

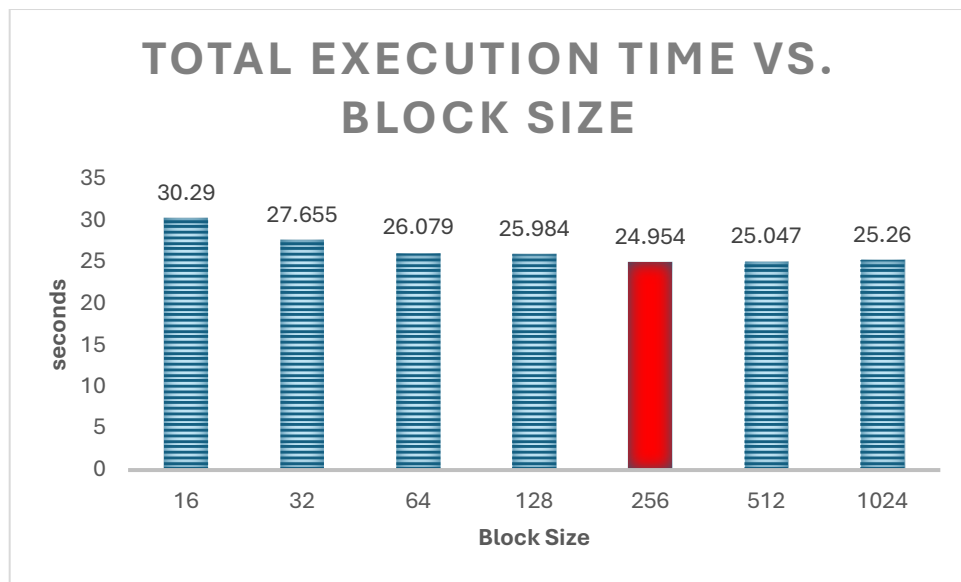
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
Running on:
Platform: QUALCOMM Snapdragon(TM)
Device: QUALCOMM Adreno(TM) 643

!!SOLUTION IS CORRECT!!

real    0m4.310s
user    0m2.258s
sys     0m0.785s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/opencl_tiled_m$
```

3.3 Neon Block Matrix Multiplication

I tested block sizes of 16, 32, 64, 128, 256, 512, and 1024 and found that performance stopped improving beyond a block size of 256.



Measurement	3.1 OpenCL Naïve MM	3.2 OpenCL Tiled MM	3.3 Neon Block MM (256x256 Block)
Real time	0m 50.718s	0m4.310s	0m24.954s
User time	0m 2.154s	0m2.258s	0m24.512s
System time	0m 0.755s	0m0.785s	0m0.323s

3.3.1 Neon Block Matrix Multiplication `time make run` output

3.3.1.1 16x16 Block Size

```
● ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make run
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda
a/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m30.290s
user    0m29.925s
sys     0m0.252s
● ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

3.3.1.2 32x32 Block Size

```
● ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include
../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m27.655s
user    0m27.105s
sys     0m0.345s
● ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```


3.3.1.3 64x64 Block Size

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lopenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m26.079s
user    0m25.573s
sys      0m0.323s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

3.3.1.4 128x128 Block Size

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lopenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lopenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m25.984s
user    0m25.549s
sys     0m0.326s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

3.3.1.5 256x256 Block Size

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m24.954s
user    0m24.512s
sys      0m0.323s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

3.3.1.6 512x512 Block Size

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make
run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpenCL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr
/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m25.047s
user    0m24.574s
sys     0m0.340s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

3.3.1.7 1024x1024 Block Size

```
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$ time make
run
cd ../../helper_lib; make
make[1]: Entering directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -c -o device.o device.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o kernel.o kernel.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o matrix.o matrix.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64
-lOpenCL -lm
gcc -g -Wall -c -o img.o img.c -I/usr/local/cuda/include -L/usr/local/cuda/lib64 -lOpen
CL -lm
ar rcs helper_lib.a device.o kernel.o matrix.o img.o
make[1]: Leaving directory '/workspaces/wes237b/assignment-3/helper_lib'
gcc -g -Wall -o solution ../../helper_lib/helper_lib.a main.c -I../../helper_lib -I/usr
/local/cuda/include ../../helper_lib/helper_lib.a -lm -L/usr/local/cuda/lib64 -lOpenCL
./solution Dataset/0/input0.raw Dataset/0/input1.raw Dataset/0/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/1/input0.raw Dataset/1/input1.raw Dataset/1/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/2/input0.raw Dataset/2/input1.raw Dataset/2/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/3/input0.raw Dataset/3/input1.raw Dataset/3/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/4/input0.raw Dataset/4/input1.raw Dataset/4/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/5/input0.raw Dataset/5/input1.raw Dataset/5/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/6/input0.raw Dataset/6/input1.raw Dataset/6/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/7/input0.raw Dataset/7/input1.raw Dataset/7/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/8/input0.raw Dataset/8/input1.raw Dataset/8/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/9/input0.raw Dataset/9/input1.raw Dataset/9/output.raw output.raw
!!SOLUTION IS CORRECT!!
./solution Dataset/10/input0.raw Dataset/10/input1.raw Dataset/10/output.raw output.raw
!!SOLUTION IS CORRECT!!

real    0m25.260s
user    0m24.806s
sys      0m0.325s
ubuntu@c20a4dfb50ef:/workspaces/wes237b/assignment-3/homework/block_mm_neon$
```

Conclusion

* Please use the ``time`` command to compare this against your implementations in Assignment 2. Is this faster for all matrices? If not, why?

Answer: No. The above we showed time outputs and will go more into details, but the takeaway is the OpenCL is faster overall but for smaller datasets, e.g. dataset0, it is slower since the overhead associated with launching the OpenCL kernel and transferring data between the host and GPU can be larger relative to the actual computation time.

* Talk about latency of this method vs neon vs plain CPU. Are there times where you might want to use different methods?

Measurement	3.1 OpenCL Naïve MM	3.2 OpenCL Tiled MM	3.3 Neon Block MM (256x256 Block)	Assignment#2
Real time	0m 50.718s	0m4.310s	0m24.954s	0m48.313s
User time	0m 2.154s	0m2.258s	0m24.512s	0m46.917s

OpenCL has higher latency due to kernel launches and data transfers between the CPU and GPU, but its parallelism makes it much faster for large matrices. NEON has lower latency because it runs SIMD instructions directly on the CPU, making it better for smaller workloads. Plain CPU has little overhead but less parallelism. Therefore, CPU(NEON) can be better for small matrices, while OpenCL is better for large matrices.

Works Cited

- [1] V. Blanco, "Intro to Practical SIMD for Graphics," [Online]. Available: https://vkguide.dev/docs/extra-chapter/intro_to_simd/.
- [2] ARM, "Getting Started with Neon Intrinsics on Android," [Online]. Available: <https://www.arm.com/technologies/neon>.
- [3] ARM, "Intrinsics," [Online]. Available: [https://developer.arm.com/architectures/instruction-sets/intrinsics/#f:@navigationhierarchiessimdisa=\[Neon\]&q=vaddvq_s32](https://developer.arm.com/architectures/instruction-sets/intrinsics/#f:@navigationhierarchiessimdisa=[Neon]&q=vaddvq_s32).
- [4] Microsoft, "ARM intrinsics," [Online]. Available: <https://learn.microsoft.com/en-us/cpp/intrinsics/arm-intrinsics?view=msvc-170>.